



THE UNIVERSITY OF
NOTRE DAME
A U S T R A L I A

MED 4000

WEEK 25

TUTOR GUIDE

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The cases and materials in this guide were reviewed by;

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RELEVANT PRIOR LEARNING:

MED 1000

- 1.03 - Human Form, Structure & Sexual Characteristics [L]
- 1.25 – History and examination of the Rheumatological system [L & CCS]
- 1.26 - Assessment of gait [L]

MED 2000

- 2.10 – Thromboembolic disease [L]
- 2.24 – Assessment and management of bony lumps [L]
 - Bone tumours [L]

MED 3000

- 3.03 – Assessment of the post operative patient [Short case tutorial]
- 3.08 – Acute torticollis [short case tutorial]
- 3.19 – Bony lumps [short case tutorial]
- 3.19 – Joint pain and swelling [short case tutorial]
- 3.22 – Rheumatoid arthritis [short case tutorial]
- 3.27 – Deep venous thrombosis [short case tutorial]

WEEK 25_SURGERY_ORTHOPAEDICS		
Yr	Wk	LEARNING OBJECTIVES:
4	25	• Know the changes in posture and gait associated with leg length discrepancies. • Outline the four phases of normal gait. • Briefly describe the common underlying causes of abnormal gaits. • Outline the factors contributing to postural abnormalities.
4	25	• List the five important post operative complications common to orthopaedic patients and briefly outline the preventative measures that can be taken with each of them. • Outline the risk factors associated with venous thromboembolism (VTE). • Know the most effective anticoagulants recommended for prophylaxis in high risk patients and explain briefly the roles, if any, of mechanical prophylaxis and aspirin play in prophylaxis. • Know the pharmacological treatment of VTE and duration of treatment in provoked versus unprovoked VTE.
4	25	• Know the four key complications associated with bone metastases in cancer patients. • Describe briefly the diagnostic techniques used in detection of bone metastases. • Outline the different modalities of treatment used in complications associated with bone metastases.
4	25	• Know how to look up whether a particular “new” drug is a) approved for registration in Australia and b) if it is listed on the PBS. • Outline the surgical treatments available to rheumatoid arthritis.
4	25	• Describe torticollis and explain its natural history • Outline the management of patients with acute torticollis including preventative measures. • List the conditions which should be considered in the differential diagnoses of a patient presenting with a wry neck.

CASE ONE

Short case number: 4_25_01

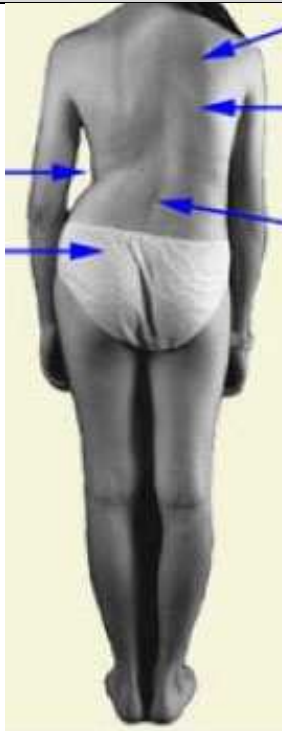
Category: Musculoskeletal system and skin

Discipline: Surgery_orthopaedics

Setting: General practice

Topic: Disorders of posture and gait

CASE



Amanda is a 14 year old high school student who presents with her mother because she has been having problems at her new school. She was constantly being teased about the way she walks. They have just moved from another state and this is creating difficulties with her settling in at the new school.

On further questioning her mother tells you that they always have to get new shoes for Amanda because one shoe always wears out faster than the other. Amanda herself said that she always has to take up one leg of her jeans more than the other. Her mother has never taken any notice of the way Amanda walked because they think it's normal. But lately since this issue came up, her mother has been paying more attention and has noticed that Amanda tends to go up and down while walking and swings one of her legs more than the other.

On examination you note that Amanda's left leg is longer than the right and her hips are tilted to accommodate the discrepancy in length. There is compensatory downward tilt of the shoulder on the left side and scoliosis of the lumbar spine.

QUESTIONS

1. Describe the common signs associated with leg length discrepancies and describe how the gait may be affected.
2. How is gait assessed? Describe the phases of normal gait and outline the other possible causes of abnormal gaits.
3. What are the three tasks of postural control? Outline the factors contributing to postural abnormalities.

Resources:

1. Henry MM and Thompson JN. Clinical Surgery, 2nd edition. Chapter 33, Principles of orthopaedics, p 645. Saunders Elsevier 2005.
2. Sweeting K. Gait and posture. Australian Family Physician, June 2007, Vol 36(6):398-405.
3. How to treat: Postural abnormalities in childhood. Australian doctor, March 2003, VI.

ANSWERS

Note additional tips from Dr Brett Courtenay, orthopaedic surgeon, St Vincent's hospital, Darlinghurst.

Question 1

Describe the common signs associated with leg discrepancies and describe how the gait may be affected.

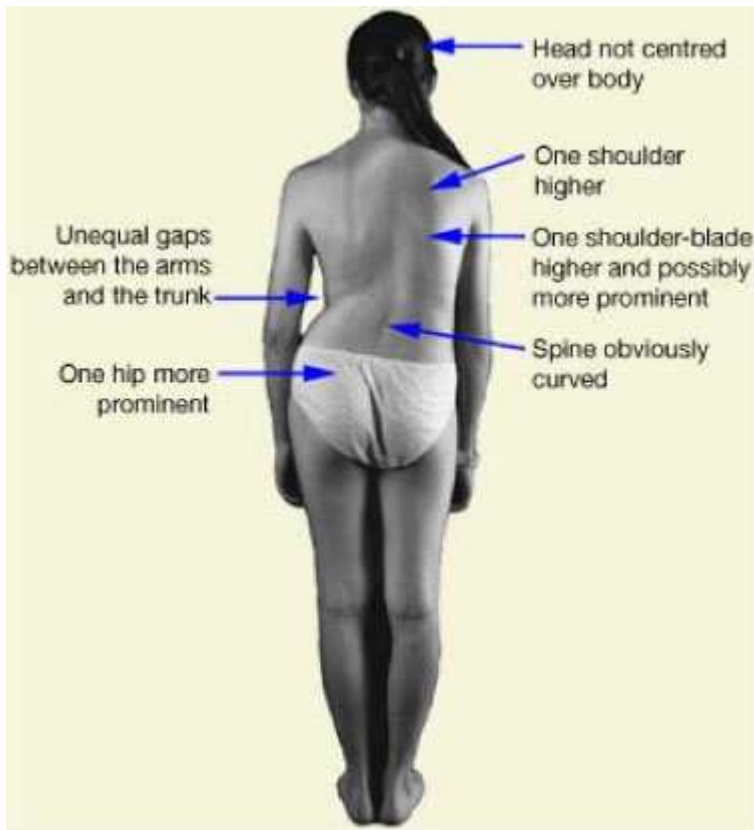


Table 1. Common signs of leg length discrepancies seen in gait

Body part	Gait manifestation
Shoulder	Dropped on the longer side
Arm	Abducted arm swing on the longer side
Pelvis	Higher on the longer side (especially noticeable in swing phase)
Leg	Circumduction on the longer side in swing
Knee	Excessive flexion on the longer side in swing to allow ground clearance
Foot	Out toe on the shorter side

Question 2

How is gait assessed? Describe the phases of normal gait and outline the other possible causes of abnormal gaits.

Often the best time to examine a patient's gait is as they are entering the examination room as they are less likely to act unnaturally. *Subtle changes are better detected with patient bending forward and seen from behind.* More thorough gait assessments can be done with the patient walking up and down a walkway of at least 6 metres or on a treadmill.

When assessing gait, it is important to understand that a problem affecting one body part can lead to problems elsewhere. A deformity in one part will be compensated in another. Due to the complex interactions of human body parts in movement, there is limited research showing causation between abnormal gait and posture and musculoskeletal pain. At a time where evidence based practice is a growing trend, clinicians working predominantly in this area (e.g. podiatrists, physiotherapists and musculoskeletal physicians) must balance this with empirical evidence.

Table 33-1. Phases of normal gait

Phase	Movement
Heel strike	The heel makes contact with the ground
Stance	Weight is being transferred from the heel to the toes
Toe off	A final push is given to the foot as it leaves the ground
Swing through	The leg is brought forward with the knee slightly flexed to allow the foot to clear the ground

During normal gait, heel strike is slightly lateral to the midline of the heel and load passes down the outside of the foot and then across the foot to the great toe. Important if looking at a shoe and trying to determine normal from abnormal.

Box 33.2 Abnormal gaits

Antalgic Painful and with a short stance phase; seen in any condition of the lower leg where the pain is exacerbated by weight-bearing, e.g. osteoarthritis of the hip

Stiff leg A fused hip or knee joint causes abnormal swing-through when the pelvis has to be rotated to bring the leg through

Trendelenburg With proximal muscle weakness the pelvis on the opposite side sags during the stance phase-seen in developmental dysplasia of the hip, poliomyelitis and in osteoarthritis of the hip

Short leg During the stance phase, the short leg results in the pelvis and shoulder on the affected side sagging down

Shuffling Seen in Parkinson's disease, it has a short swing-through and no real heel strike or toe off

Stamping The swing-through phase is abnormal with a broad base and high stepping-often caused by peripheral neuropathy with tabes dorsalis

Ataxic Broad-based with unsteadiness on turning-cerebellar disease, multiple sclerosis or head injury

Foot drop During swing-through, the foot scuffs on the ground-an L5 root lesion, common peroneal nerve palsy or old poliomyelitis

Scissor Occurs in children with cerebral palsy with adductor spasm, so the swing-through of one leg is blocked by the other

Question 3

What are the three tasks of postural control? Outline the factors contributing to postural abnormalities.

There are three tasks of postural control:

- antigravity function – ie. maintenance of upright posture and keep eyes level

- maintenance of equilibrium, and
- providing mechanical support for motor action.

Optimal posture implies balanced distribution of body mass around the centre of gravity where the compression force on spinal discs is balanced by ligamentous tension and there is minimal energy expenditure from postural muscles.

A quick visual scan by the doctor will detect significant asymmetries in posture – presence of head tilt, unlevelling of shoulders, pelvis or scoliosis (back view); forward head carriage where ear lobes are anterior to shoulders, exaggeration of spinal curves (side view).

Factors contributing to postural abnormalities

- Structural changes: Congenital or acquired structural changes (eg. idiopathic scoliosis, deformed vertebra, foot and ankle abnormalities)
- Abnormal gait e.g. pes planus, leg length discrepancy
- Muscle imbalance: The commonest cause according to Janda¹ is muscle imbalance arising from modern lifestyle with constrained postures from repetitive tasks and limited range in movement and inactivity.

Postural syndromes (Janda)

‘Muscle imbalance’ refers to the tendency of certain muscles to become tight, hypertonic, and activating earlier and more readily in movement patterns whereas other muscles are more likely to exhibit hypotonia, inhibition, weakness, atrophy and activate less readily and later in movement patterns. This tendency is not random but occurs in a predictable and recognisable pattern. The tight, hypertonic muscles have predominately antigravity function during gait whereas their antagonists tend toward weakness and inhibition.

Muscle imbalances result in faulty postures, joint malalignment and development of altered inefficient movement patterns resulting in strain and degeneration of joints, myofascial and ligamentous structures. *Look out for early signs of joint damage.*

¹ Janda is a neurologist, rehabilitation physician and researcher of muscle function analysis.
Janda V. Evaluation of muscular imbalance. In: Liebensohn C, editor. Rehabilitation of the spine. Baltimore. Lippincott Williams & Wilkins 1996;97–112

CASE TWO

Short case number: 4_25_02

Category: Musculoskeletal system and skin.

Discipline: Surgery_Orthopaedics

Setting: Orthopaedic ward in a hospital

Topic: Assessment of the post operative orthopaedic patient

CASE



Jane Andrews [3_27_06], the 45 year old woman who presented to Emergency Department with swelling of her left leg and diagnosed as deep vein thrombosis (DVT) ten years ago. She is now admitted for elective surgery because of recurrent dislocation of the right patella. Although Jane has poor muscle tone generally, she is not obese, has had no major illnesses. Significant history includes bilateral tubal ligation soon after the DVT episode.

QUESTIONS

1. List the five important complications common to most orthopaedic operations and briefly outline what are the preventative measures that can be undertaken.
2. Venous thromboembolism (VTE) is a term which encompasses deep vein thrombosis (DVT) and pulmonary embolism (PE). Outline the risk factors associated with VTE and discuss this in the context of Jane's procedure.
3. The mortality rates following VTE are 5–23% but in symptomatic patients with adequate anticoagulation, mortality is reduced to 1–2%. What are the most effective anticoagulants recommended for prophylaxis in high risk patients? What roles, if any, do mechanical prophylaxis and aspirin play in prophylaxis?
4. What is the pharmacological treatment of VTE and what is the duration of treatment in provoked versus unprovoked VTEs? Briefly outline the 3 situations where LMWH are used in preference to warfarin.

Resources:

1. Solomon L, Warwick DJ and Nayagam S. Apley's concise system of orthopaedics and fractures, 3rd edition. Chapter 1, p 134-5. Hodder Arnold 2005.
2. Chapman N, Brighton T, Harris M, Caplan GA, Braithwaite J and Chong BH. Venous thromboembolism. Australian Family Physician, Jan/Feb 2009, Vol 38, No 1/2:36-40.

ANSWERS

Note additional tips from Dr Brett Courtenay, orthopaedic surgeon, St Vincent's hospital, Darlinghurst.

Question 1

List the five important complications common to most orthopaedic operations and briefly outline what are the preventative measures that can be undertaken.

The 5 important complications which are common to most orthopaedic operations are:

- **Swelling** is common after operations on the limbs. Its effects are often aggravated by tight dressings or plaster casts. Prevention is all important: 1) Bandages must be firm enough to stop swelling but not to constrict. To be effective it must go from the toes to the groin; otherwise it will cause venous compression distally and increase VTE risk; 2) the limb should be elevated; 3) movements should be encouraged; and 4) if a cast is used, it should be well padded or split.
- **Haematoma formation:** Postoperative bleeding or oozing from cut surfaces may be considerable. Haematoma formation is reduced by surgical technique which may include suction drainage (as cut bone surfaces cannot collapse under suction).
- **Delayed wound healing:** Poor peripheral circulation may cause delayed healing or breakdown of the wound. The circulation should always be checked before any operation is undertaken.
- **Infection:** Postoperative infection is always a problem but in joint replacement surgery.
- **Thromboembolism:** (next question).

Question 2

Venous thromboembolism (VTE) is a term which encompasses deep vein thrombosis (DVT) and pulmonary embolism (PE). Outline the risk factors associated with VTE and discuss this in the context of Jane's procedure.

Add smoking to list of risk factors for VTE below.

Table 1. Risk factors associated with VTE³⁴

General

- Older age
- Immobility, paresis
- Malignancy
- Obesity
- Previous VTE
- Family history of VTE
- Oral contraceptive pill, hormone replacement, tamoxifen
- Venous insufficiency/varicose veins

High risk clinical situations

- Surgery (especially hip and knee surgery or major surgery for malignancy)
- Pregnancy/puerperium
- Acute medical illness
- Congestive cardiac and respiratory failure
- Trauma
- Central venous catheter

Question 3

The mortality rates following VTE are 5–23% but in symptomatic patients with adequate anticoagulation, mortality is reduced to 1–2%. What are the most effective anticoagulants recommended for prophylaxis in high risk patients? What roles, if any, do mechanical prophylaxis and aspirin play in prophylaxis?

As most VTE is provoked by easily recognisable risk factors, effective prophylaxis offers the opportunity of major improvements in health care. Australian guidelines on VTE risk assessment and prophylaxis based on international recommendations suggest prophylaxis for hospitalised patients with an additional VTE risk factor.

The most effective anticoagulants (recommended for prophylaxis in highest risk patients) are the low molecular weight heparins (LMWH) and fondaparinux. The primary contraindication for pharmacological prophylaxis is a high risk of bleeding or a hypersensitivity to a specific anticoagulant that may result in an adverse reaction.

Indication for therapy	Medication*
Prevention of VTE	
• Orthopaedic surgery – elective total hip replacement – elective total knee arthroplasty – hip fracture surgery • Major surgery (intra-abdominal or surgery lasting >45 minutes) and age >40 years • Medical Illness with additional risk factors[†] • Pregnancy and postpartum + additional risk factors • Travel >5000 km[‡]	• LMWH/fondaparinux + GCS/IPC • LMWH/fondaparinux + GCS/IPC • LMWH/fondaparinux + GCS/IPC • LMWH/UFH/fondaparinux + GCS/IPC • LMWH/UFH/fondaparinux • LMWH • Adequate hydration, exercise, avoid alcohol, nonrestrictive clothing

Mechanical prophylaxis, (ie. intermittent pneumatic compression [IPC] stockings or graduated compression stockings [GCS]) is recommended for patients with a higher than normal risk of bleeding or as an adjunct to more efficacious pharmacological prophylaxis.

Mechanical means include calf compressors, worth a mention. Stockings have recently been getting some bad hits as actually increasing the risk.

It should be noted that **aspirin**, although more effective than placebo in preventing VTE in orthopaedic patients, is less effective than other thromboprophylactic agents and can cause bleeding.

Given the efficacy of prophylaxis, clear guidelines for hospital patients, and known mortality rate of VTE, uptake of prophylaxis is still surprisingly poor. Hospital audits have found that only 72% (in Australia) of high risk surgical patients and 42% (in Australia) of eligible medical patients received VTE prophylaxis.

Note from Dr Brett Courtenay:

The most contentious issue is with this lady is how long she is immobilized either completely in a cast or brace which will occur and then partially being until she can get sufficient muscle action to actually effectively work her deep veins.

*The follow on is how long does she get prophylaxis, given she cannot get calf compression or other non chemical means. **There is no answer to this and no guidelines at present for post discharge.***

Question 4

What is the pharmacological treatment of VTE and what is the duration of treatment in provoked vs. unprovoked VTEs? Briefly outline the 3 situations where LMWH are used in preference to warfarin.

Chemical prophylaxis now includes some additional oral medications which selectively affect parts of the intrinsic pathways and are becoming very popular. They are worth a mention.

The pharmacological treatment for VTE consists of:

- initiating immediate acting subcutaneous LMWH or unfractionated heparin or fondaparinux (subcutaneous) with oral warfarin and
- continuing the injected drug for at least 5 days and until international normalised ratio (INR) is therapeutic (2.5 ± 0.5).

The therapeutic effectiveness and safety of warfarin relies on maintaining the INR in the therapeutic range. A high variability in dose response between patients and significant interactions with drugs and diet means that warfarin therapy is difficult and time consuming to manage. Nonetheless, it remains the most effective and cost efficient drug currently available for the treatment of the majority of patients with VTE.

With the comment about warfarin being the most effective and cost efficient drug it is worth mentioning it is also potentially the most dangerous. I would hate to think that students were not aware of the risks.

- LMWH may be preferred to warfarin in some patients, including **cancer patients** for whom treatment with warfarin is associated with an increased risk of bleeding and VTE recurrence
- LMWH are also recommended over warfarin for long term therapy in **pregnant women** (due to the teratogenic effects associated with warfarin), and
- Bridging therapy in **patients scheduled to undergo surgery** (due to LMWH's shorter half life therapeutic anticoagulation can continue closer to surgery time).

Whatever the drug being prescribed, Australian guidelines recommend continuation of treatment for **at least 3 months for provoked VTE** and **6-12 months for unprovoked VTE**. In some cases, treatment may be continued indefinitely. Treatment decisions depend on the patient's predisposition to VTE (provoked or unprovoked), the site and extent of thrombus, the presence or absence of symptomatic embolism, and bleeding risk.

Note from Dr Brett Courtenay:

My own approach would be to try and work out why she got the DVT 10 years ago, was it the pill or some other factor. With the tubal ligation did she get much prophylaxis and did she have a problem. Also I would use LMWH and Doppler scan her at 7 to 10 days when the sutures are coming out and if DVT then either continue the LMWH for the period until she was walking well. If a very big clot then warfarin for 3 - 6 months.

CASE THREE

Short case number: 4_25_03

Category: Musculoskeletal system and skin

Discipline: Surgery_Orthopaedics

Setting: General practice

Topic: Bone tumours and complications

CASE

George O'Connor is a 49-year old roofing contractor, who presented with back pain in Med2000 Week 5 and was subsequently diagnosed as multiple myeloma. This time he presents a year later with recent onset of severe pain over the lower lumbar spine and weakness in his legs.



Past history: lumbosacral spine radiographs show osteoporosis and anterior wedging/crush fractures. He had radiotherapy to the crush fracture at T8 and this gave him rapid pain relief. He underwent aggressive chemotherapy for multiple myeloma and was successfully treated with iV pamidronate for hypercalcemia.

The last time you had contact with him was "... George's serum paraprotein concentration level is very low, and he has no pain. He is on a disability pension and feels well, although he has no illusions about the eventual outcome... He has become an avid reader of medical literature, and frequently asks about new treatment methods and drugs."

Despite the fact that he has neither bladder nor bowel symptoms, you recognise the seriousness of his symptoms and the risk of spinal cord compression. So you immediately arrange for an ambulance to transport him to the hospital.

QUESTIONS

1. Outline the 4 key complications associated with bone metastases in cancer patients.
2. What diagnostic techniques are used in detection of bone metastases?
3. Describe the different modalities of treatment used in complications associated with bone metastases.

Resources:

1. Henry MM and Thompson JN. Clinical Surgery, 2nd edition. Chapter 33, Principles of orthopaedics, p 652-4. Saunders Elsevier 2005.
2. Trahair T. How to treat: Cancer in children. August 2008, p 31-38.

Question 1

Outline the 4 key complications associated with bone metastases in cancer patients.

Bone metastases are a common manifestation of distant relapse from many types of solid cancers, especially those arising in the lung, breast and prostate. As many as 80% of patients with solid cancers develop painful bone metastases to the spine, pelvis, and extremities during the course of their disease. They most commonly affect the axial skeleton, and this pattern of distribution suggests that physical properties of the circulation within the bone marrow cavity, including capillary structure and the sluggish blood flow, assist in the establishment of metastases. Bone metastases are typically referred to as 'lytic', 'sclerotic' or 'mixed', according to the radiographic appearance of the lesions.

In this case scenario, although multiple myeloma is not an example of secondary bone metastases, nevertheless the management principles are the same. MM is typically characterized by a marked increase in osteoclast activity and proliferation. Bone radiographs in multiple myeloma patients typically have a "moth-eaten" appearance due to the multiple lytic lesions.

The considerable morbidity of bone metastases includes pain, impaired mobility, hypercalcaemia, pathological fracture, spinal cord or nerve root compression and bone marrow infiltration.

Note from Brett Courtenay: Hypercalcaemia with metastases is a very bad prognostic sign. In fact if unable to be medically managed surgery usually leads to death.

a) Bone pain

Metastases usually occur in the axial skeleton. A useful rule of thumb is that metastases are unlikely below the elbows and below the knees, as there is no bone marrow in these sites. Pain from skeletal metastases may be worse at night or on weight-bearing. If complicated by fracture there may be increased pain on movement. Nerve impingement may result in a dermatomal or peripheral nerve distribution of pain.

Spinal instability is a cause of back pain in around 10% of patients with metastatic bone disease. This can cause excruciating pain, which is mechanical in origin. Because the pain is due to spinal instability, radiotherapy or systemic treatment will not relieve the pain.

In peripheral bone metastases, there may be a place for prophylactic pinning of long bones if there is risk of fracture. The risk factors are: lytic metastases, large size, cortical involvement, location such as femoral neck. Surgery is also required for mechanically unstable spine or obtaining tissue diagnosis for unknown primary.

The key to successful rehabilitation is early diagnosis, high-dose corticosteroids, rapid assessment and urgent referral for both decompression and spinal stabilization or radiotherapy.

b) Hypercalcaemia is probably the most common metabolic complication of malignant disease and is clinically important because of the associated morbidity. If left untreated, moderate to severe hypercalcaemia (serum calcium >3.0 mmol/l) causes a number of unpleasant symptoms related to dysfunction of the gastrointestinal tract, kidneys and central nervous system. With even

higher serum calcium levels, renal function and conscious level deteriorate, and death ultimately follows as the result of cardiac arrhythmias and acute renal failure.

c) Pathological fracture: Metastatic destruction of bone reduces its loadbearing capabilities, resulting initially in trabecular disruption and microfractures, and subsequently total loss of bony integrity. Rib fractures and vertebral collapses are most common. These result in loss of height and, when severe and multiple, lead to kypho-scoliosis and a degree of restrictive lung disease. However, fracture of a long bone or epidural extension of tumour into the spine causes the most disability.

d) Spinal cord compression

- Pain precedes neurological dysfunction and is the most common presenting symptom therefore requiring a high index of suspicion.
- Bladder and bowel symptoms must be considered. They are often subtle and usually predate the more obvious radicular symptoms.
- Consider this in patients with spinal metastases who develop weakness or numbness of limbs or sudden increase in back pain. Diagnosis is by CT and/or MRI and treatment is urgent.
- Give high-dose steroids **immediately** and obtain a surgical opinion, as a recent randomised trial has demonstrated better functional outcomes if surgery is possible.
- Radiotherapy is an alternative treatment and can be given postoperatively also.

Patients with axial skeletal pain and/or radicular symptoms should be assessed to exclude impending spinal cord compression, which could lead to paraplegia, urinary retention and faecal incontinence. A sensory level and increased reflexes below the level of compression can normally be identified. An MRI of the **whole** spine (to also exclude other sites of cord compression) is the investigation of choice.

Neurological recovery is unlikely if the spinal compression is not relieved within 24–48 h.

Most important prognostic factor is **ambulatory status**. If paralysis >48 hours, little chance of recovery, should only receive radiotherapy for palliation of pain. if <48 hours, 10-20% of recovery, if previously ambulatory, 75% remain ambulatory.

Question 2

What diagnostic techniques are used in detection of bone metastases?

- **Radioisotope bone scan** is the most sensitive method for detecting bony metastases – which are usually distributed in the axial rather than the appendicular skeleton.
- CT with bone windows may be more useful in defining anatomy, particularly in the spine, where MRI may also be useful.

Question 3

Describe the different modalities of treatment used in complications associated with bone metastases.

The goals of palliative treatment of bone metastases are pain relief, preservation of function, and maintenance of skeletal integrity.

- Local field **external beam radiation therapy** (EBRT) is one well-recognized and effective palliative modality. When bone pain is limited to a single or a limited number of sites, EBRT to a local field can provide pain relief in 80-90% of cases, with complete pain response obtained in 50-60%. Although treatment can be effective for patients with mild, moderate, or severe

pain, early intervention may be useful in maintaining quality of life and minimizing side effects of analgesic medications. For the majority of patients, EBRT provides excellent palliation for localized metastatic bone pain. In most clinical situations this can be achieved with a short treatment schedule of 1-5 fractions.

- **Radiopharmaceuticals** are now available for the palliation of metastatic bone pain. Strontium-89 has been shown to be as effective as wide-field radiotherapy in prostate cancer and, because of the preferential uptake of strontium at sites of new bone formation, is probably most effective for sclerotic metastases. Samarium-153 is also preferentially taken up at sites of bone formation, and is suitable for outpatient use. Recent data indicate that the agent has a significant effect on bone pain and analgesic consumption.
- **Vertebroplasty/cementoplasty** is a therapeutic alternative for the treatment of pain associated with vertebral body compression fractures. involves the injection of artificial bone cement and an opacifier into the inter-trabecular marrow space of the fractured vertebra. It is performed with imaging guidance under local anaesthesia and intravenous conscious sedation, or rarely under general anaesthesia. The procedure takes 1–2 hours, depending on the number of vertebrae requiring treatment.

A large calibre needle is placed into the involved vertebral body and radio opaque bone cement (eg. polymethyl methacrylate) is injected. The injected bone cement does not re-expand the collapsed vertebra; it acts as an internal splint to reinforce and stabilise the fracture for pain alleviation.

Table 1. Indications and contraindications for vertebroplasty^{18,19}

Indications	Contraindications	
	Absolute	Relative
Osteoporotic vertebral compression fracture(s) causing severe pain and unresponsive to conservative therapy	Fracture that is healed or is responding to conservative management	Fracture >4 months, unless nonunion has occurred
Painful vertebral compression fractures(s) caused by metastases or multiple myeloma, may or may not be receiving adjuvant radiation, surgical therapy or chemotherapy	Absence of bone oedema or gas cleft on MRI Presence of untreated coagulopathy Presence of local infection	Loss of integrity of the posterior wall of the vertebral body

Vertebroplasty provides

- significant reduction in analgesia use and offers long term pain relief.
- increases patient mobility.
- is a minimally invasive day stay procedure with a low incidence of complications.

In September 2005, the Minister for Health and Ageing approved the Medical Services Advisory

Committee's recommendation for Medicare Benefits Schedule rebates for vertebroplasty for patients suffering from painful osteoporotic vertebral compression fractures not controlled by conservative medical therapy, and for patients with pain from metastatic deposits or multiple myeloma in a vertebral body.

CASE FOUR

Short case number: 4_25_04

Category: Musculoskeletal system and skin

Discipline: Surgery_Orthopaedics

Setting: General practice

Topic: Rheumatoid arthritis_surgical management

CASE

Felicity Masters [3_22_1], the 35 year old who presented to the practice with early symptoms and signs of rheumatoid arthritis is now 12 years older and presenting because of worsening pain and deformity of her left hand despite being on methotrexate. Fortunately her right hand, although painful and swollen at times, is still functional.



Her husband, Fred, was retrenched a couple of years ago and has been unable to find suitable work. Being the sole breadwinner, she is conscious of the need to keep her hands in the best shape possible given that she is a physiotherapist.

Felicity asks for help in finding the best treatment option. She has done a lot of research herself and brought along information about a new drug and also wants to discuss surgical options.

QUESTIONS

1. When asked about a particular “new” drug which you are not familiar with, how do you go about finding out a) if it is approved for registration in Australia and b) if it is listed on the PBS and, if applicable c) what are the recommendations of the peak body representing that particular specialty for its use?
2. Outline how you would discuss the surgical treatments available to patients with severe deformities due to rheumatoid arthritis.

Resources:

1. <http://www.tga.gov.au/docs/html/medregs.htm>
2. <http://www.medicareaustralia.gov.au/provider/pbs/drugs2/rheumatoid.jsp>
3. Biggs A. Overview of the Pharmaceutical Benefits Scheme. Updated Jan 2003. <http://www.aph.gov.au/library/intguide/sp/pbs.htm>
4. Australian Rheumatology Association: Updated Recommendations for the use of biological agents for the treatment of rheumatic diseases. <http://www.rheumatology.org.au/otherpages/biological-guidelines.asp>
5. Henry MM and Thompson JN. Clinical Surgery, 2nd edition. Chapter 33, Principles of orthopaedics: Arthritis, p 648-50. Saunders Elsevier 2005.

ANSWERS

Question 1

When asked about a particular “new” drug which you are not familiar with, how do you go about finding out a) if it is approved for registration in Australia and b) if it is listed on the PBS and, if applicable c) what are the recommendations of the peak body representing that particular specialty for its use?

a) <http://www.tga.gov.au/docs/html/medregs.htm>

The lawful supply of any therapeutic good in Australia requires that the product is included on the Australian Register of Therapeutic Goods (ARTG). There are two types of entry on the ARTG. Some products are 'registered' and have an AUST R number on their label. Other products are 'listed' and have an AUST L number. the TGA decides if products are suitable for listing or require registration. Manufacturers are not permitted to suggest or imply in their advertisements that inclusion in the ARTG is a recommendation or endorsement by the TGA.

b) Checking on PBS listing status and the restrictions on use:

<http://www.medicareaustralia.gov.au/provider/pbs/drugs2/rheumatoid.jsp>

c) **Australian Rheumatology Association: Updated Recommendations for the use of biological agents for the treatment of rheumatic diseases.**

<http://www.rheumatology.org.au/otherpages/biological-guidelines.asp>

Question 2

Outline how you would discuss the surgical treatments available to patients with severe deformities due to rheumatoid arthritis.

When conservative measures have failed, surgical options need to be considered.

Synovectomy : This does not work if there is joint damage so some surgeons would perform an MRI to detect if there is joint damage. If joint damage a Synovectomy does not help! Surgical excision of the synovium or tendon sheath in rheumatoid arthritis may be beneficial. In the past it was done as an open procedure with consequent considerable joint morbidity, but this has been replaced by arthroscopic synovectomy with special powered instruments.

Arthroscopy : A joint that is swollen and inflamed will usually benefit from an arthroscopic washout which removes inflammatory mediators and fragments of cartilage. However it has no long lasting effects and would rarely be used alone.

Joint surgery: Long-term or permanent relief of the symptoms of arthritis requires surgery on the bones of the affected joint. A variety of options are available and the choice depends on a number of factors, including:

- Age
- expectations of the patient
- occupation
- the joint affected.

Table 33-4. Surgical options for an arthritic joint

Procedure	Effects	Example
Arthrodesis	Stiff but pain-free joint	Fusion of the lumbar spine or ankle
Excision	Removal of one aspect of the joint may relieve pain; shortening of the limb beyond the resection	Keller's operation on the big toe for hallux valgus
Osteotomy	Alteration of the line of load transmission to an unaffected part of the joint	High tibial osteotomy for unicompartmental osteoarthritis of the knee
Prosthetic replacement	Excision of diseased joint surfaces and replacement of surfaces with metal and high-density polyethylene	Hip, knee and shoulder

Table 33-5. Types of joint replacement

Type	Nature	Comments	Examples
Constrained	Simple hinge	Loosening because of inevitable rotational movement	Elbow, wrist
Semi-constrained	Simple hinge, but some rotation possible	Still subject to loosening	Elbow, wrist
Unconstrained	Two independent parts so that stability depends on sound anchorage in bone and the soft tissues	Most common type in use; highly satisfactory	Hip, knee, ankle, shoulder, elbow

In the knee, the replacement can be unicompartmental if one side only is affected; more usually, all three compartments are replaced.

page 650

Almost any joint in the arm or leg can be replaced. Usually the replacement consists of a metal component bearing on a high-density polyethylene surface. The prosthesis may be cemented into place with methyl-methacrylate bone cement or be uncemented, with the surface textured to encourage bone in growth. New surfaces include the use of hydroxyapatite (HA), which is the basic mineral of bone. With HA chemically bonded to the surface of the prosthesis, the patient's bone can directly bind the implant.

The first modern replacement was the hip joint developed by Charnley in 1961, although previous attempts had been made to replace the hip with wood or ivory. The modern hip joint can be expected to last for 15-20 years but this depends upon:

- surgical experience
- state of the recipient bone
- prosthesis design
- stresses placed upon the implant

- the materials used at the bearing surfaces-recent research has supported hard-on-hard articulation with ceramic-on-ceramic or metal-on-metal.

Complications of joint replacement surgery include:

- general complications of any major operation
- specific complications of the procedure, including intraoperative fractures, postoperative dislocations and fractures, infection of the implant and loosening.

Loosening may be secondary to infection or an aseptic mechanical process. If the prosthesis is mechanically loose, it can be removed and replaced, but when infection is present, the safest option is to remove all the foreign material, identify the infective agent and treat this vigorously. At a second procedure, it may then be possible to insert another prosthesis.

Joint replacement surgery in rheumatoid arthritis carries particular risks. The immune response is altered, and infection rates are higher. At operation, the bones are often osteoporotic with an increased risk of intraoperative fractures. When multiple joints are involved, the surgeon may be embarking on an extended programme of replacement.

The problem with joint replacements today in younger patients is more about loosening of the implants at the bone implant junction due to excessive loads. There is still a small wear issue but not as much as before.

Note from Brett Courtenay:

The problem with joint replacements today in younger patients is more about loosening of the implants at the bone implant junction due to excessive loads. There is still a small wear issue but not as much as before.

Rheumatoid patients because of their lower demands tend to do much better. Also because all of their articular cartilage is affected they do not do well with osteotomies or fusions as it puts too much load on other damaged joints.

Orthopaedic surgeons would consider that in a rheumatoid patient coming to surgery there is at least some damage to all joints. The results of further replacements after initial failure (revision surgery) are not as good as those of the primary procedure, and with each subsequent attempt the surgery becomes more challenging.

CASE FIVE

Short case number: 4_25_05

Category: Musculoskeletal system and skin

Discipline: Surgery

Setting: General practice

Topic: Acute torticollis [SDL]

CASE

Kylie is a 25 year old graphics designer who presents with sudden onset of pain and stiffness on the left side of the neck since the early hours of this morning. She had no fever, headache or other symptoms.



On further questioning, Kylie tells you that she had been working pretty long hours at her computer prior to a launch. At work she is usually stuck on her computer screen for hours at a time and lately she hasn't had time to have a break. In the past she has had occasional stiffness in her neck on one side or other but this is the first time it has been so severe as to stop her from driving.

QUESTIONS

1. Outline the information you would give to Kylie on torticollis, its natural history and management.
2. What are the conditions that should be considered in the differential diagnoses for neck pain?

Resources:

1. Creswick M. How to treat: Musculoskeletal neck disorders. *Australian Doctor*, October 2004, p 27-34.
2. Murtagh J. Patient information: Acute wry neck. *Australian Doctor*, 2003.

Question 1

Outline the information you would give to Kylie on torticollis, its natural history and management.

Torticollis/"wry neck" is neck muscle spasm usually accompanied by pain. Acute torticollis occurs typically in young people between 12-30 years of age. Chronic torticollis, which is rare, occurs mainly in middle-aged adults.

The main symptom is severe pain localised to the middle or, more usually, on one side of the neck. The pain can be referred to the head or shoulder region. The neck feels "stuck" in one position and any attempted movement to free it usually results in sharp spasms of pain.

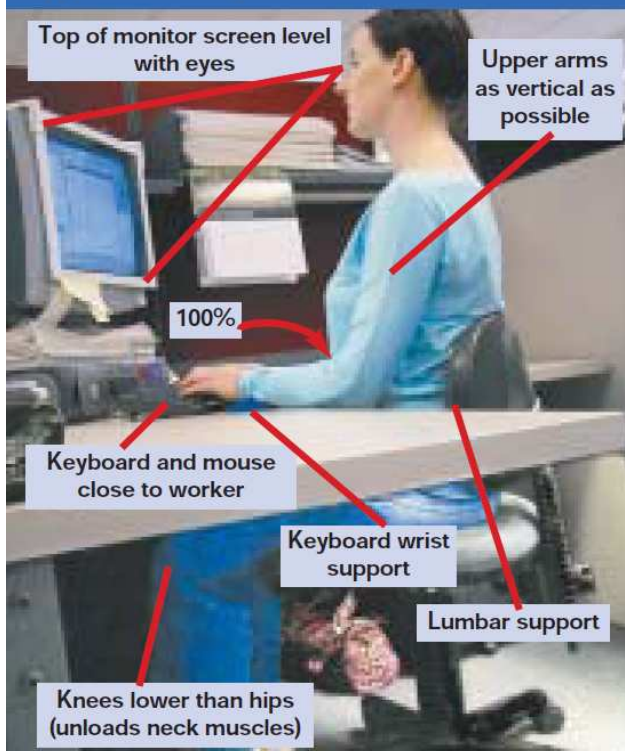
Contrary to popular belief, a wry neck is not caused by a cold draught on the neck. The mechanism is most likely due to injury to one or more of the facet joints resulting in the joints acting as though they were jammed/locked. This results in painful, protective muscle spasm.

Fortunately an acute wry neck is a transient and self-limiting condition that can recover in 24 hours. Sometimes the problem can last for about a week. There are usually no risks or complications. However, neck problems can return.

A variety of treatments are offered from several different schools of therapy but it is important to remember that most cases get better quickly without special treatment, so conservative management is advisable.

- **Analgesics** such as paracetamol and aspirin for 2-3 days
- **Heat and massage:** Applying a cold pack or heat in the form of a hot face washer or heat pack to the painful area. Massage using an analgesic balm can also relieve symptoms.
- **Sleeping and pillows:** Sleep with the painful side on a low, firm pillow and avoid sleeping on too many pillows.
- **Exercises:** Keep your neck mobile within your comfort zone. Gentle exercise aids a more rapid recovery.
- **Posture:** Keep a good posture, including keeping your chin tucked in and your neck upright in a vertical position for tasks such as reading and writing.
- **Driving** during the initial period of a wry neck should be avoided because the inability to get full rotation of the head to view oncoming traffic is hazardous.
- **Collars** can provide support and comfort for a short period for acutely painful necks but they are not generally recommended. The neck is best kept mobile and exercised naturally by normal use. However, if the pain with movement is severe, wearing a support for a few days will help but, if possible, avoid wearing it during sleep.
- **Spinal manipulation** is inadvisable because of protective muscle spasm. However, some physical treatments which provide excellent relief include gentle mobilisation and a therapy called muscle energy therapy.

Figure 7: A desirable, balanced, sustained seated posture that is strain-free. (A slouched posture shortens anterior chest muscles and places sustained load on the posterior neck muscles to support the head.)



Prevention: If the patient sits at a desk at work, advise sitting with knees slightly lower than the hips to unload the spinal muscles, including the dorsal neck muscles (figure 7). Head sets should be used for prolonged telephone work. Use of a shaped, rolled edge pillow to support the head in bed at night is supported by a Cochrane review. Mini-breaks from the workstation posture may help.

Question 2

What are the conditions that should be considered in the differential diagnoses for neck pain?

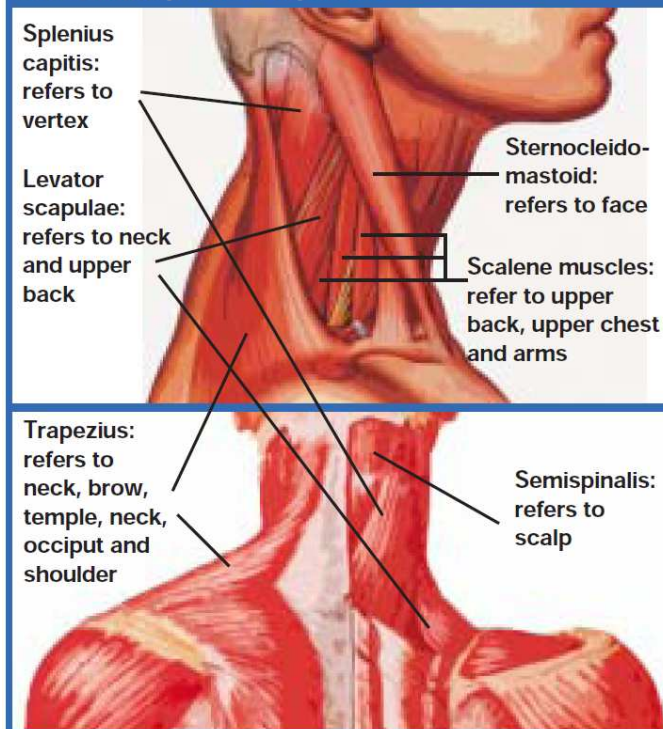
Neck pain can have many causes, including:

- Degenerative joint disease (not in Kylie, unless she has had a history of trauma)
- Neuronal compression by spinal neurocentral canal or lateral foramen stenoses.
- Spinal instability (from trauma or inflammatory disease such as rheumatoid arthritis), inflammatory joint disease (rheumatoid arthritis, psoriatic arthritis).
- Torticollis.
- Referred visceral pain (heart, gall bladder) and local visceral pathology (tumour or infection).

MUSCULOSKELETAL neck pain may originate from:

- Musculo-tendinous tender points at the attachment of cervical muscles to the skull.
- The cervical spine, from intervertebral or segmental dysfunction, cervical spondylosis, arthritis of the atlantooccipital joint or disc annulus tears.
- Myofascial pain referred from the muscles of the neck, that is, trigger points.
- Lateral foraminal stenoses by disc or bone, causing radicular pain.
- Entrapment neuropathy of the greater or lesser occipital nerve.

Figure 4: Neck muscles involved in head and neck pain and their common pain referral patterns.



A careful history, including a history of conditions such as rheumatoid arthritis, psoriasis, colitis and family history, is very important. The examination should be directed by knowledge of common musculoskeletal pain generators and their patterns of referral, and should always include palpation of the facet joint column and large muscles of the neck.

Patterns of pain referral:

The upper neck refers pain to the scalp and upper face. The mid-neck causes local neck pain or subclavicular pain, and the lower neck refers to the upper back, shoulders and arms. In patients with neck pain, always ask about headache, arm or upper back symptoms, and vertigo